

TORONTO METROPOLITAN UNIVERSITY

CIND 110

DATA ORGANIZATION FOR DATA ANALYSTS

Assignment 2

This is an individual, scenario-based assignment designed to test your understanding of data organization, hierarchical modeling, and text mining. It consists of **four parts**: (I) XML architecture and querying, (II) JSON backend migration, (III) Information Retrieval (IR) analytics, and (IV) self-assessment and perspective.

Academic Integrity and Use of Generative AI (Policy 60)

You may use **GenAI** to help brainstorm schema fields, draft dummy data, and generate the raw XML dataset.

All BaseX querying, JSON conversion/modeling, MongoDB queries, and Python/IPYNB Information Retrieval implementations must be your own work and reviewed by you (no GenAI-written scripts for the core tasks).

Scenario

- You must build a functioning data pipeline and analysis prototype that meets the needs of three distinct business teams.
- You are the Lead Data Architect for a Botanical Research Consortium.
- You are producing this for the *Inventory Operations team* who needs reliable XML reporting on stock, the *Web Development team* who needs a modern JSON backend for their app, and the *Research Analytics team* who needs to mine flower descriptions to find similarities.
- Your final product must be a cohesive set of deliverables that act not just as code repositories, but as a technical briefing justifying your design decisions to these specific teams.
- Your work will be evaluated on technical insight, the strength of your reasoning and justifications, the effectiveness of your solutions, and mature self-assessment of the system's limitations.

Dataset Requirements

- **Source File:** flowerdb.xml
- **Required Entities:** *Flowers, Families, Shops, Inventory, Customers, Orders.*
- **Recommended Size:** ~20 flowers, ~10 families, ~5 shops, ~20 inventory rows, ~12 customers. Increase counts if needed so that each query returns **at least one result**.
- **Family descriptions:** Ensure the description text is **at least 15 words** to support the Part III IR task.
- **Minimum Required Fields:**
Family(familyId, familyName, description);
Flower(flowerId, name, familyId, seasons e.g., ["Spring", "Summer"]);
Shop(shopId, name, city);
Inventory(shopId, flowerId, quantity);
Customer(customerId, name, city).
- You must include your prompts in the Appendix if GenAI produced the file.

Part I: XML Architecture & Querying (Operations Team)

Goal: The Inventory Operations team needs specific reports extracted from your XML hierarchy. Execute XPath or XQuery/FLWOR in **BaseX** on flowerdb.xml.

I.A Queries Write and execute queries to answer the following business questions:

1. Flowers that bloom in **Summer**. (*Output: list distinct flowerId, name*).
2. Customers in the same city as shop locations. (*Output: list distinct customerId, name, city*).
3. Flowers that are **never** stocked in any shop. (*Output: list distinct flowerId, name*).

I.B Aggregation/Grouping

4. Flowers that bloom in **Spring** but are **not** in the *Asteraceae* family (*or another single family of your choice; state which*). (*Output: flowerId, name, familyName*).
5. For each **family**, show its name and the **total stock quantity** across all shops. (*Output: familyName, totalStock*).

Deliverables (Part I)

- Your flowerdb.xml file.
- A markdown/document containing query code, **BaseX screenshots** of results, and a brief **justification** of your XML element/path design choices.
- **Identify which fields might pose data privacy risks** (e.g., Customer data) and briefly state how the Operations team should handle them.

Part II: JSON Backend Migration (Web Dev Team)

Goal: The Web Development team requires a JSON backend. Convert the XML into JSON and run JSON queries using MongoDB (or equivalent) to ensure parity with the old system.

II.A XML $\Rightarrow$ JSON Modeling

- Convert `flowerdb.xml` into **JSON**; either as *one consolidated file* or a *set of files*.
- Keep clear, consistent **IDs** and cross-references so relationships can be joined.
- Choose to **embed** related data or use **references**.

II.B Querying JSON

- Re-implement in JSON the **same five query intents from Part I**.
- Use the equivalent filters/joins/aggregations needed for your JSON design (e.g., using MongoDB `$lookup`).

Deliverables (Part II)

- `flowerdb.json` *or* a set of JSON files (pretty-printed).
- Query code & screenshots demonstrating that each JSON query successfully matches its XML counterpart.
- A markdown cell/note **justifying** one major modeling choice you made (e.g., embedding vs. referencing) to optimize performance for the Web team.

Part III: Information Retrieval Pipeline (Analytics Team)

Goal: The Research Analytics team wants to find relationships between plant families based on text descriptions. Complete an IR workflow in **Jupyter (IPYNB)**.

III.A Extract & Prep From `flowerdb.xml`, extract `FamilyName` and `Description` into a two-column CSV using BaseX.

III.B IPYNB Text Mining Using Python in your notebook:

1. Apply at least **three** text pre-processing steps (e.g., lowercasing, stop-word removal, lemmatization).
2. Create a **unigram** Term–Document Matrix (TDM); display the dimensions.
3. Create a **bigram** TDM; display the dimensions.
4. Compute **cosine similarity** and identify families most similar to *Apocynaceae* (or another target family).

Deliverables (Part III)

- Your extracted CSV file.
- Your **IR IPYNB** notebook with code and outputs visible.
- Markdown cell: **Justify** your text mining approach to the Analytics team. Which preprocessing step most changed the similarity? When were bigrams more informative?

Part IV: Self-Assessment & Perspective

Goal: Demonstrate **self-knowledge** and **perspective** by critiquing your own design, evaluating its real-world viability, and reflecting on your learning.

IV.A Critical Reflection Critique your dataset and pipeline design by addressing the following areas:

- **Data Limitations:** What are the limits, flaws, or biases of your synthetic dataset?
- **Ethics & Privacy:** What data privacy and ethical concerns would you need to address before this pipeline could process real customers' and shops' data?
- **Stakeholder Impact:** How might these dataset biases and privacy concerns impact the business teams' day-to-day work if this prototype were deployed?
- **Technical Reflection:** Knowing what you now know about hierarchical modeling and IR, what would you do differently if you had to build this pipeline again?

Deliverables (Part IV)

- A formal postscript Markdown cell in your notebook (or attached document) answering the four reflection areas outlined above.

Submission Guidelines

Submission Checklist

- `XML_Data_StudentName.DOCX` (*Code, BaseX screenshots, Part I notes*)
- `JSON_Data_StudentName.DOCX` (*Queries, screenshots, Part II notes*)
- `IR_StudentName.ipynb` (*Notebook with outputs, Part III/IV markdown*)
- Include source data: `flowerdb.xml`, `flowerdb.json`, and the IR CSV.
- **Disclose and cite** any GenAI assistance in an Appendix.

Grading Rubric

Criteria	Points
Pipeline demonstrates an insightful use of XML, JSON, and Python structures. Assesses BaseX querying, MongoDB schema design, and IR implementation, specifically looking at how well the code handles unique structures and edge cases of the generated dataset.	20
Design choices (data privacy, embedding vs. referencing, text preprocessing) are justified with thorough arguments. Recommendations must be supported by specific, concrete evidence drawn directly from the generated flowerdb data rather than generic theoretical statements.	20
XML queries, JSON models, and text mining outputs successfully resolve the distinct problems of the Operations, Web, and Analytics teams. Translates technical outputs (like cosine similarity scores) into clear, actionable business insights for non-technical stakeholders.	20
Code is highly readable, excellently commented, and functionally accurate end-to-end. The deliverables transparently document the technical process, including how specific challenges were addressed.	20
Reflection demonstrates sound self-knowledge by identifying the limits, flaws, and real-world ethical/privacy implications of the pipeline. Includes a critical evaluation of the assumptions, biases, or structural errors present in the GenAI output used to generate the initial XML data.	20
Total	100